



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)
BSc CHEMISTRY**

Programme	B.Sc Chemistry				
Course Title	INORGANIC CHEMISTRY-III				
Type of Course	MAJOR				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Different segments of environment, Environmental pollution, Different types of pollution, health effect and hazards associated with chemicals, Acid-base concepts, basic idea about chemical analysis				
Course Summary	This course explains different types of environmental pollution, its causes, consequences, remedies of prevention and develop concerns for the environment It reveals the importance of need of green chemistry and green synthesis It gives an idea about major acid base concepts and reactions in nonaqueous solvents It initiates the students for exploitation of advanced materials in the demand of changing trends of modern industry. This course explores students to the role and opportunities of Chemistry as a discipline in the modern era and develops skills in qualitative and quantitative analysis of inorganic compounds.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To critically evaluate issues in environment	An	C	Instructor-created exams / Assignments
CO2	To gain insights into the economic and environmental aspects of green chemistry and to contribute the advancement of sustainable practices in the field of chemistry	Ap	C	Assignment / seminar/quizzes
CO3	To understand the theories of acids and bases and reactions in non aqueous solvents and to identify compounds as acids and bases	Ap	C	Assignment/Seminar/Class test
CO4	To understand and apply principles of material chemistry and its facets	Ap	C	Assignment/Seminar/Class test
CO5	To equip the students with familiarization in separation and identification of ions	Ap	P	Group work /Assignment/class test/
CO6	Practicum in inorganic qualitative analysis with hands on familiarity with various ions	An	P	Group work /Assignment/ Viva, Observation of practical skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Mark
I	ENVIRONMENTAL CHEMISTRY AND GREEN CHEMISTRY		15	33

	1	Environment and Environmental pollution, pollutant and contaminant, Segments of Environment	1	
	2	Air Pollution : Types of air pollutants, Gaseous air pollutants (oxides of carbon, nitrogen and sulphur), Particulates.	2	
	3	Effects of air pollution: Smog (London and Los Angeles Smog), Global warming, ozone depletion, acid Rain,	2	
	4	Control of air Pollution, alternative refrigerants	1	
	5	Water Pollution : Sources of water pollution, Eutrophication, Bioaccumulation, bioMagnification,	2	
	6	Water quality parameters (DO,BOD and COD and their determination), Toxic metals in water(Pb, Cd and Mg), control of water pollution	2	
	7	Soil pollution, Thermal pollution, noise pollution, light pollution, radiation pollution (Sources and effects)	3	
	8	Introduction to Green chemistry – Goals of Green chemistry, Twelve basic principles of green chemistry- Atom economy- Green solvents- water, supercritical fluids, ionic liquids [mention with examples).	2	
II	ACID BASE CONCEPTS AND NON-AQUEOUS SOLVENTS		5	10
	9	Major acid-base concepts, Arrhenius, Bronsted-Lowry, Solvent system, Lux-Flood, Lewis and Usanovich concepts.	2	
	10	Non - aqueous Solvents: Classification – General Properties	1	
	11	Reactions in Liquid Ammonia, liquid SO ₂ and Liquid HF	2	
III	MATERIAL CHEMISTRY		15	33
	12	Introduction and scope of material chemistry-	1	
	13	Ceramic materials: Definition, classification(traditional and advanced ceramics) , Composition- oxides and nitrides- general	5	

		properties- applications- structural, Electronics, thermal and biomedical applications		
	14	Catalytic materials : zeolites, alumina- surface properties, supporting materials	3	
	15	Composite materials: Definition and types of composite materials (polymer matrix composites, metal matrix composites, carbon matrix composites- explanation with examples	3	
	16	Inorganic solids :Perovskites- ABX_3 - $CaTiO_3$, $LaCoO_3$, spinel compounds- AB_2O_4 - $MgAl_2O_4$	3	
IV	INORGANIC QUALITATIVE AND QUANTITATIVE ANALYSIS		10	22
	17	Inorganic qualitative analysis: Need for elimination of interfering acid radicals and their elimination methods – oxalate, fluoride, borate, phosphate, chromate, arsenite and arsenate	1	
	18	Principles of separation of basic radicals into various groups – Solubility product and Common Ion effect – Their application in the qualitative inorganic analysis.	2	
	19	Micro analysis: merits and application, preparation of sodium carbonate extract and its merits	1	
	20	Inorganic Quantitative Analysis – Gravimetric analysis – Introduction – Types of gravimetric analysis, Precipitation, Advantages and disadvantages of gravimetric analysis –	2	
	21	Properties of precipitates and precipitating agents, Mechanism of precipitate formation- Von Weimarn equation and its applications– Co-Precipitation and post precipitation –	2	
	22	Homogeneous and heterogeneous precipitation – gravimetric factor - Inorganic and Organic precipitating agents and their applications - NH_3 , H_2SO_4 , $NH_4 SCN$, oxine, cupron, cupferron, 1-nitroso-naphthol, dithiocarbamates.	2	

V	INORGANIC CHEMISTRY PRACTICAL III:INORGANIC QUALITATIVE AND QUANTITATIVE ANALYSIS		30	
		1. Inorganic qualitative analysis: a) Study of reactions of following ions, <i>Anions:</i> carbonate, sulphate, fluoride, chloride, acetate, borate, oxalate, phosphate and nitrate <i>Cations:</i> Lead, bismuth, copper, cadmium, iron, aluminium, cobalt, nickel manganese, zinc, barium, calcium, strontium, magnesium and ammonium b) Systematic analysis of mixtures containing two cations and two anions from the above list (Na ₂ CO ₃ extract procedure may be adopted)		
		2. Inorganic quantitative analysis : Gravimetric analysis (Open ended) a) Estimation of barium as barium sulphate or sulphate as barium sulphate can be done		

References:

1. A.K. De, Environmental Chemistry, 6th Edn., New Age International Pvt. Ltd., New Delhi, 2006.
2. S. S. Dara, A Textbook of Environmental Chemistry and Pollution Control, 8th Edn., S. Chand and Sons, New Delhi, 2008.
3. K. Ahluwalia, Environmental Chemistry, Ane Books India, New Delhi, 2008.
4. Paul T. Anastas, T. C. Williamson, Green Chemistry – Designing Chemistry for the Environment, 2nd Edn., 1998.
5. S. C. Ameta, R. Ameta, Green Chemistry: Fundamentals and Applications, CRC Press, 2013.

6. Anthony R. West, Solid State Chemistry and its Applications, 2 nd Edn., Wiley-Blackwell, 2014.
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8. N.N. Greenwood and A.Earnshaw, *Chemistry of Elements*, 2/e, Elsevier Butterworth-Heinemann, 2005.
9. J.E.Huheey, E.A.Keiter, R.L.Keiter. O.K.Medhi. *Inorganic Chemistry, principles of structure and reactivity*, Pearson Education, 2006.
10. G.L.Miessler, D.A.Tarr, *Inorganic Chemistry*, Pearson, 2010.
11. M.J. Starfield and Shrager, *Introductory Material Science*, McGraw Hill.
12. K.K. Chowla, *Composite Materials*, Springer – Verlag, NY,1987.
- 13.. M. Tinkham, *Introduction to Superconductivity*, McGraw Hill, 1975.
14. A.V. Narlikar and S.N. Edbote, *Superconductivity and Superconducting Materials*, South Asian Publishers, New Delhi, 1983.
15. G. Svehla, *Vogel's Qualitative Inorganic Analysis*, 7th Edn., Prentice Hall, New Delhi, 1996.
16. J. Mendham, R. C. Denney, J. D. Barnes, M. Thomas, *Vogel's Textbook of Quantitative Chemical Analysis*, 6th Edn., Pearson Education, Noida, 2013.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	-	1	2	1	2	3		1	1	2	3	2
CO 2	2	1	3	2	2	3	2		1		1	3	2
CO 3	3	1		2	2	3	3		1	1	1	1	2
CO 4	3	-	2		1	3	3		2	1	2	1	2
CO 5	2		2	-	2	2	2		2		2	2	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment /viva/seminar	Practical skill evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	